

Data Card: EncodeLang African Language Corpus

Project: Bridging the Tokenization Gap in African LLMs via SuperBPE

Dataset: Team Krugger African Language Corpus

Initial languages: Yoruba, Igbo, and Swahili

Input vs. Output

What are users inputting? What are the outputs or labels your model will predict or generate?

The dataset contains primarily text from three languages: **Yoruba, Igbo, and Swahili**. The dataset is used to train and evaluate tokenization systems rather than to generate language translations. The output is a sequence of tokens produced from the input text. English translations are not required for the core tokenization task, although parallel English data could be incorporated in future work to support additional linguistic and semantic analysis.

Classification Problem

What is the classification task, if any? What categories or classes are you working with, and why?

This is not primarily a classification problem. The dataset is intended for unsupervised tokenizer training and evaluation. The main comparison examines how Standard BPE and SuperBPE segment African-language text and how many tokens they require to represent the same text. Analysis will be conducted separately for Yoruba, Igbo, and Swahili to identify differences in tokenization behaviour across the languages.

Labeling

Where do your labels come from? How will you check for consistency and accuracy?

The core dataset does not require manually assigned labels because BPE-based tokenization learns from statistical patterns within the text. Each document can be associated with metadata such as language, source, and dataset split. Language identification will be checked where possible, while duplicates, corrupted records, and incorrectly identified language data will be removed or flagged. Where appropriate, samples may be reviewed using reliable linguistic resources or by fluent speakers.

Data Creation

How will you collect your data? Will it come from direct sources or be synthesized?

The dataset will be assembled from existing African-language corpora and other legally usable text resources containing Yoruba, Igbo, and Swahili. One of the primary sources used in the

current dataset is a publicly available dataset hosted on Mendeley Data. Sources are selected based on relevance, language coverage, quality, licensing, and diversity of content. The collected text is cleaned and standardized while preserving meaningful linguistic patterns. The resulting corpus will be divided into training, validation, and evaluation sets.

Web Collection Ethics

Do you plan to source data from the web? If yes, which sites or platforms? What tools will you use? Have you addressed consent, copyright, and privacy?

Yes. The current dataset includes data obtained from a publicly available dataset hosted on **Mendeley Data**:

<https://data.mendeley.com/datasets/compare/jccjsk6pd3>

No external AI tools were used to generate the dataset. The data was collected, inspected, and cleaned for use in the tokenization research. For future sources, we will review licensing and permitted use before incorporation, document data provenance, and avoid intentionally collecting private or personally identifiable information. Public availability will not be treated as automatic permission for unrestricted reuse.

Bias and Representation

Who might be left out of your dataset? How will you ensure that the dataset reflects community diversity?

Africa has thousands of languages, while this project initially focuses on only three: **Igbo, Yoruba, and Swahili**. Consequently, many African languages, linguistic communities, dialects, and writing varieties are not represented. The current selection reflects the scope and resources available for this initial research rather than the relative importance of these languages. Future work can expand the dataset to additional African languages and improve representation across linguistic and regional communities.

Values and Community Alignment

How does your dataset design reflect your values? How will the community benefit?

The dataset is guided by **linguistic independence, cultural preservation, fairness, community representation, transparency, and responsible data stewardship**. The project seeks to reduce the dependence of language technology on resources designed primarily around high-resource languages and investigate whether African languages can be represented more effectively within

LLM infrastructure. Transparent documentation of data sources, limitations, and representation supports accountability while contributing to research that can strengthen the representation of African languages in AI systems.